

AviLight Ecological Impact and Diagnostics Report

Executive Summary

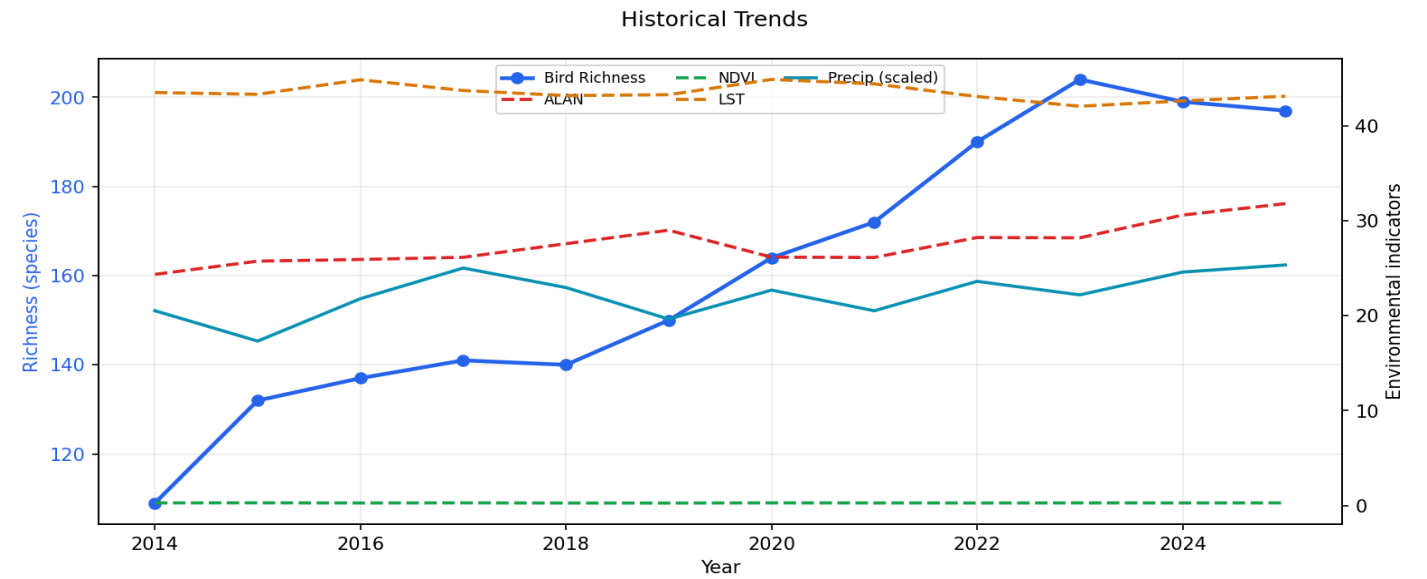
Run Context

Field	Value
Area	All Areas
Trend Years	2014-2025
Snapshot	2025-01
Generated	2026-04-19 16:11:24 UTC

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Ecological Impact View

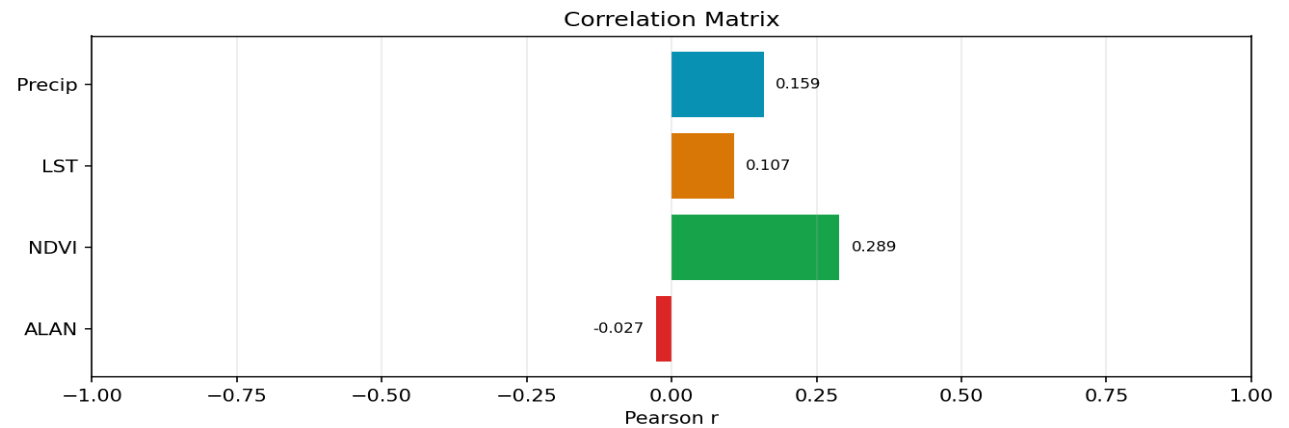
Chart: Long-Term Ecological Impact Trends



Numerical Values: Historical Trends

Year	Richness	ALAN	NDVI	LST	Precip
2014	109.00	24.35	0.2770	43.51	2052.59
2015	132.00	25.75	0.2874	43.32	1732.52
2016	137.00	25.92	0.2711	44.86	2179.36
2017	141.00	26.15	0.2829	43.73	2502.38
2018	140.00	27.58	0.2659	43.19	2296.71
2019	150.00	29.01	0.2643	43.28	1964.26
2020	164.00	26.17	0.2801	44.89	2269.63
2021	172.00	26.14	0.2759	44.42	2052.01
2022	190.00	28.24	0.2698	43.09	2361.96
2023	204.00	28.21	0.2786	42.06	2219.41
2024	199.00	30.60	0.2762	42.63	2459.51
2025	197.00	31.81	0.2805	43.12	2534.40

Chart: Correlation Matrix

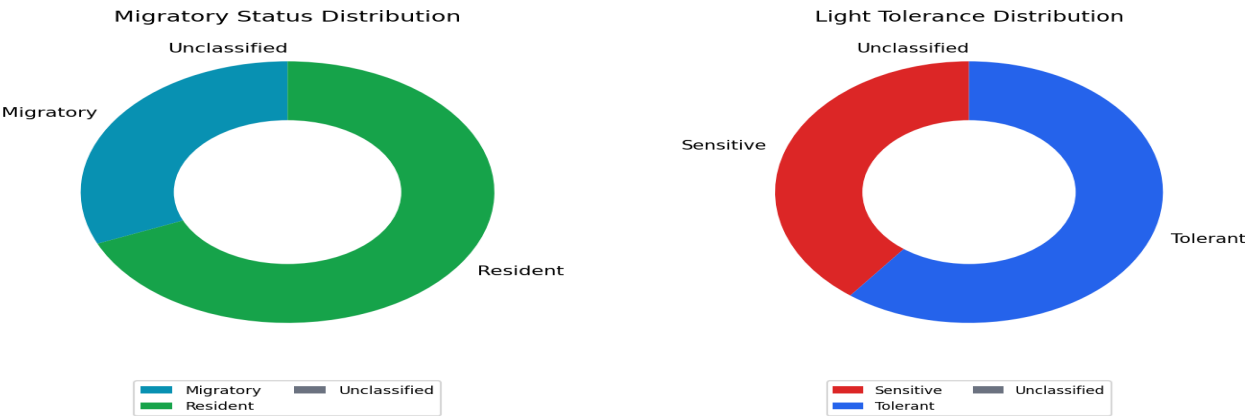


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Numerical Values: Correlation Coefficients

Metric	Pearson r
ALAN vs Richness	-0.0271
NDVI vs Richness	0.2895
LST vs Richness	0.1075
Precipitation vs Richness	0.1587

Chart: Migratory and Light Tolerance Distribution



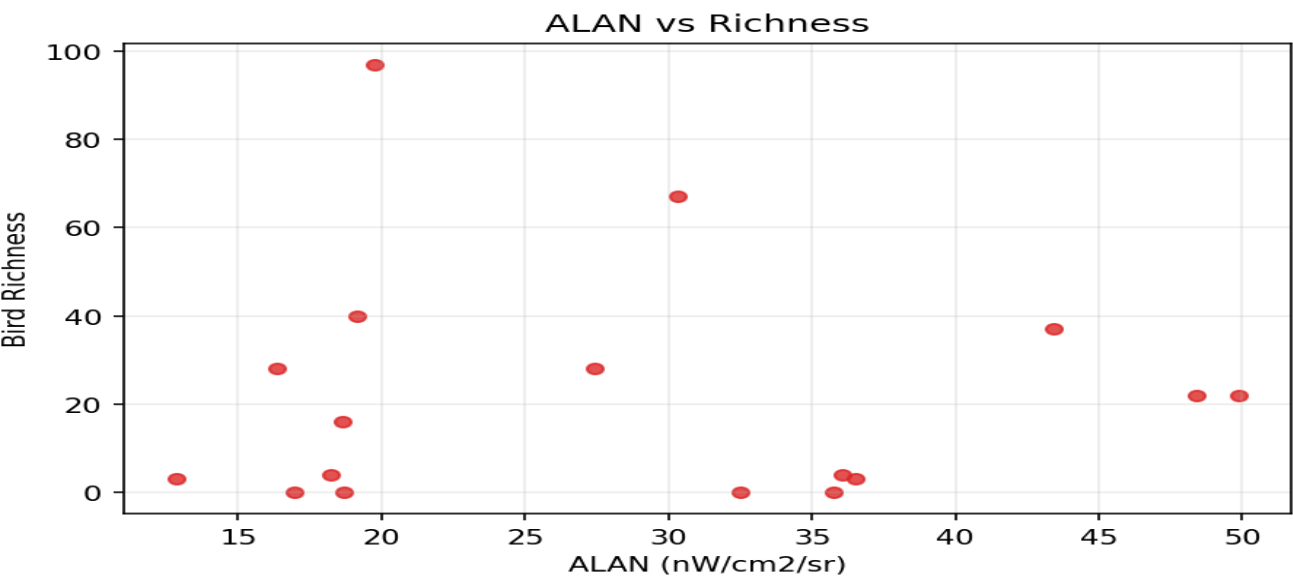
Numerical Values: Migration Distribution

Category	Count
Migratory	39
Resident	85
Unclassified	0

Numerical Values: Light Tolerance Distribution

Category	Count
Sensitive	49
Tolerant	75
Unclassified	0

Chart: ALAN vs Richness

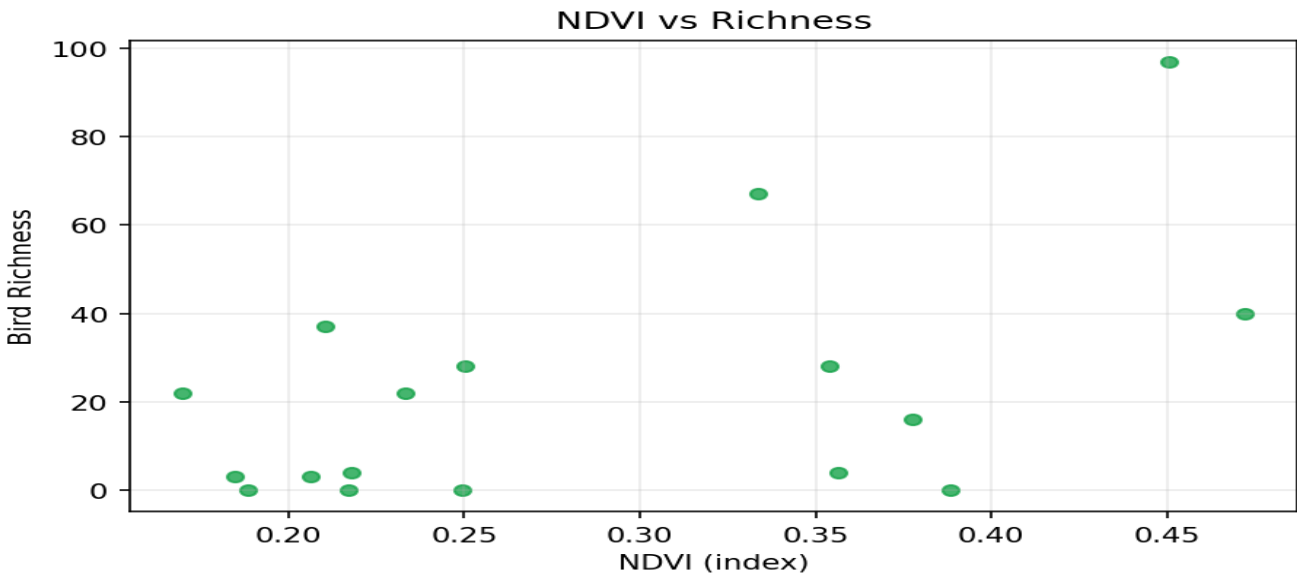


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Numerical Values: ALAN vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	16.9669	0.00
Las Piñas	Las Piñas	18.6431	16.00
Makati	Makati	43.4287	37.00
Malabon	Malabon	18.6865	0.00
Mandaluyong	Mandaluyong	36.0768	4.00
Manila	Manila	48.4198	22.00
Marikina	Marikina	18.2037	4.00
Muntinlupa	Muntinlupa	19.1179	40.00
Navotas	Navotas	12.8473	3.00
Parañaque	Parañaque	30.3321	67.00
Pasay	Pasay	49.8610	22.00
Pasig	Pasig	35.7396	0.00
Pateros	Pateros	32.5225	0.00
Quezon City	Quezon City	19.7503	97.00
San Juan	San Juan	36.5149	3.00
Taguig	Taguig	27.4281	28.00
Valenzuela	Valenzuela	16.3417	28.00

Chart: NDVI vs Richness



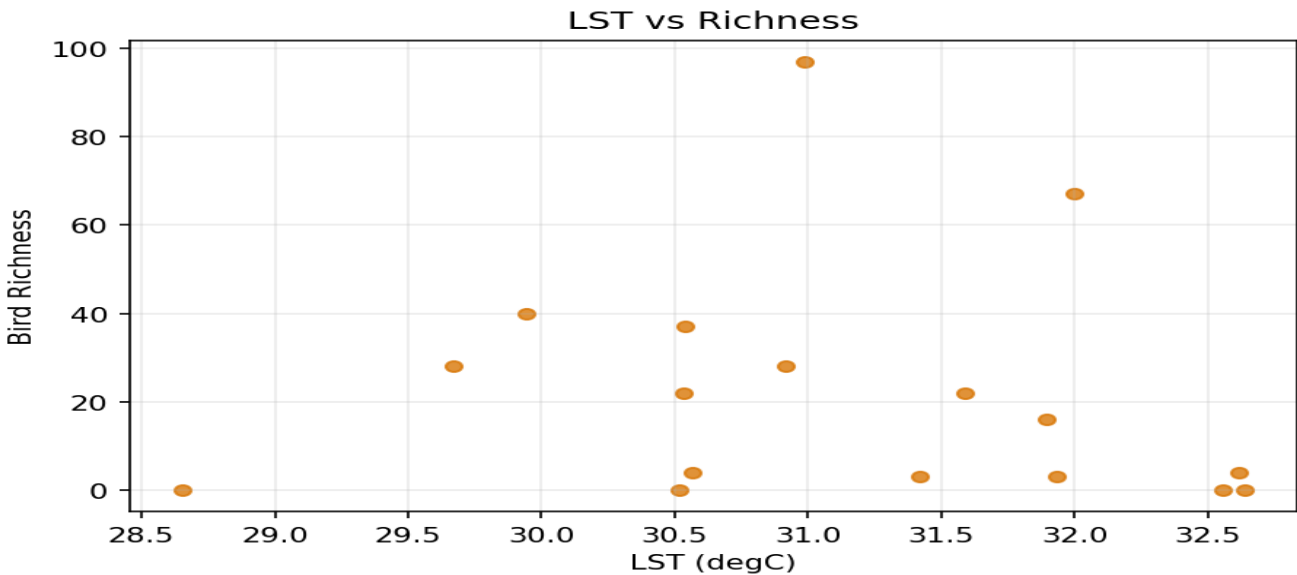
Numerical Values: NDVI vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	0.3884	0.00
Las Piñas	Las Piñas	0.3776	16.00
Makati	Makati	0.2104	37.00
Malabon	Malabon	0.1886	0.00
Mandaluyong	Mandaluyong	0.2179	4.00
Manila	Manila	0.1699	22.00
Marikina	Marikina	0.3565	4.00
Muntinlupa	Muntinlupa	0.4719	40.00
Navotas	Navotas	0.1848	3.00

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Parañaque	Parañaque	0.3337	67.00
Pasay	Pasay	0.2332	22.00
Pasig	Pasig	0.2496	0.00
Pateros	Pateros	0.2173	0.00
Quezon City	Quezon City	0.4504	97.00
San Juan	San Juan	0.2063	3.00
Taguig	Taguig	0.3540	28.00
Valenzuela	Valenzuela	0.2501	28.00

Chart: LST vs Richness

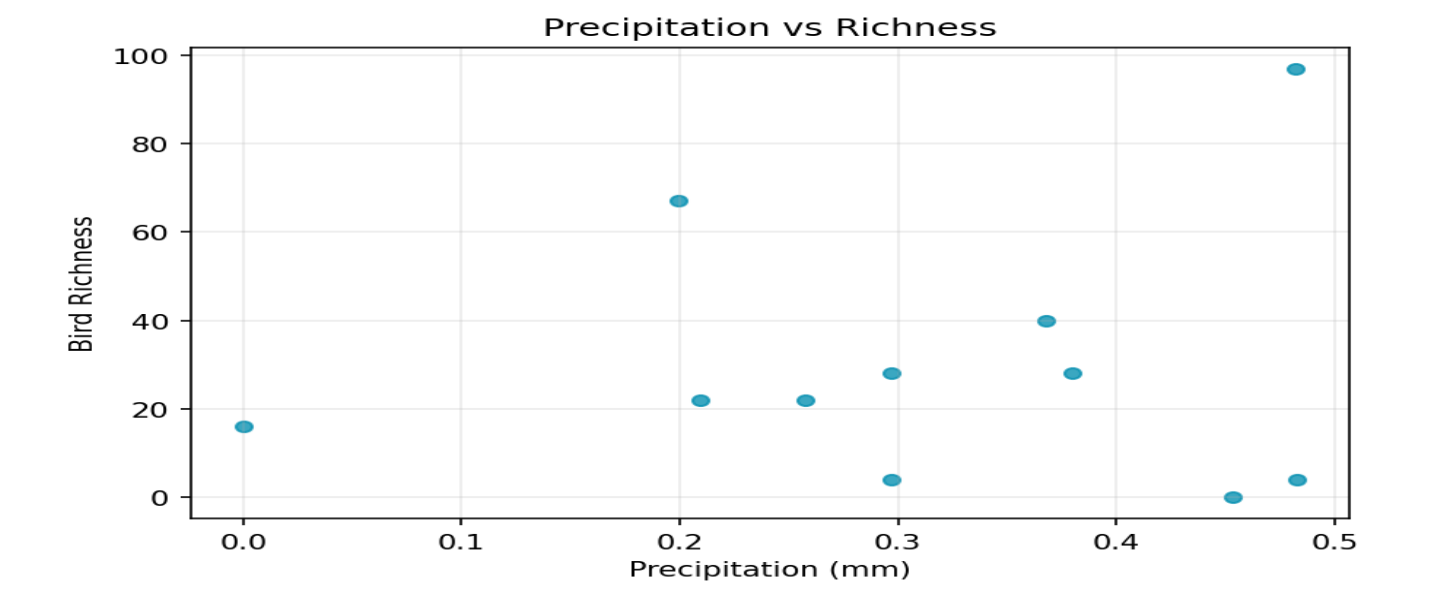


Numerical Values: LST vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	30.5167	0.00
Las Piñas	Las Piñas	31.8972	16.00
Makati	Makati	30.5417	37.00
Malabon	Malabon	28.6532	0.00
Mandaluyong	Mandaluyong	32.6184	4.00
Manila	Manila	30.5324	22.00
Marikina	Marikina	30.5682	4.00
Muntinlupa	Muntinlupa	29.9428	40.00
Navotas	Navotas	31.4188	3.00
Parañaque	Parañaque	31.9967	67.00
Pasay	Pasay	31.5880	22.00
Pasig	Pasig	32.6377	0.00
Pateros	Pateros	32.5588	0.00
Quezon City	Quezon City	30.9879	97.00
San Juan	San Juan	31.9312	3.00
Taguig	Taguig	30.9138	28.00
Valenzuela	Valenzuela	29.6679	28.00

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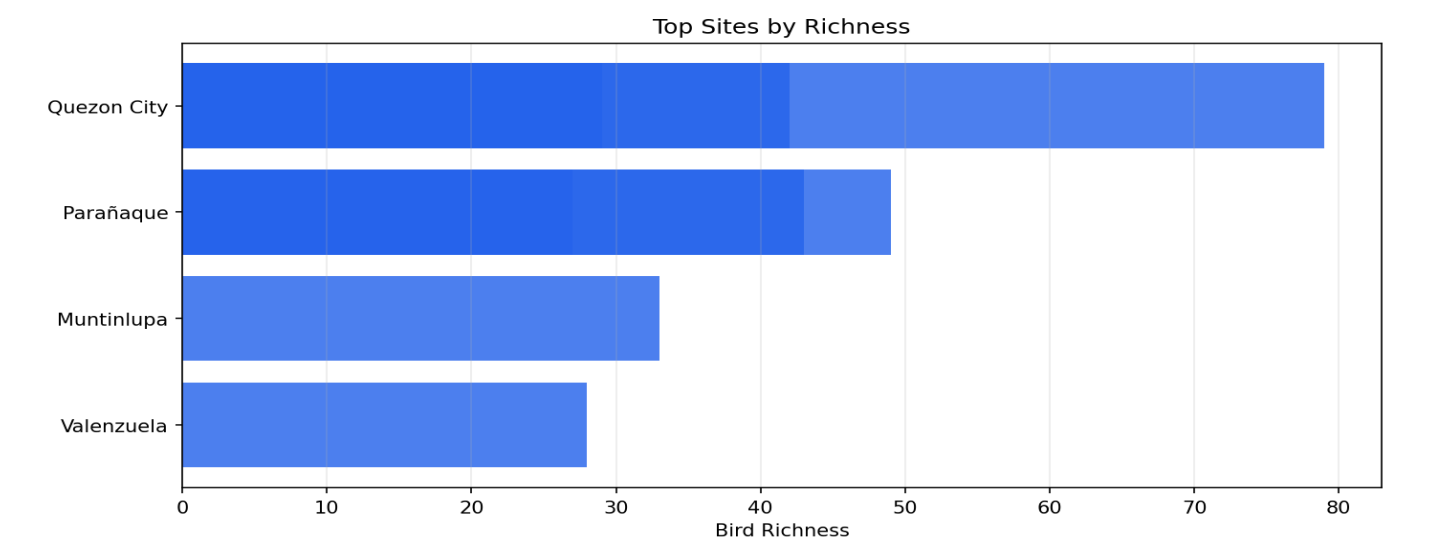
Chart: Precipitation vs Richness



Numerical Values: Precipitation vs Richness

Site	Area	X	Richness
Las Piñas	Las Piñas	0.0000	16.00
Mandaluyong	Mandaluyong	0.2972	4.00
Manila	Manila	0.2094	22.00
Marikina	Marikina	0.4827	4.00
Muntinlupa	Muntinlupa	0.3678	40.00
Parañaque	Parañaque	0.1991	67.00
Pasay	Pasay	0.2575	22.00
Pasig	Pasig	0.4534	0.00
Quezon City	Quezon City	0.4818	97.00
Taguig	Taguig	0.3800	28.00
Valenzuela	Valenzuela	0.2970	28.00

Chart: Top 10 Sites by Richness



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Numerical Values: Top Sites by Richness

Rank	Area	Richness
1	Quezon City	79
2	Parañaque	49
3	Parañaque	43
4	Quezon City	42
5	Muntinlupa	33
6	Quezon City	29
7	Quezon City	29
8	Valenzuela	28
9	Parañaque	27
10	Quezon City	26

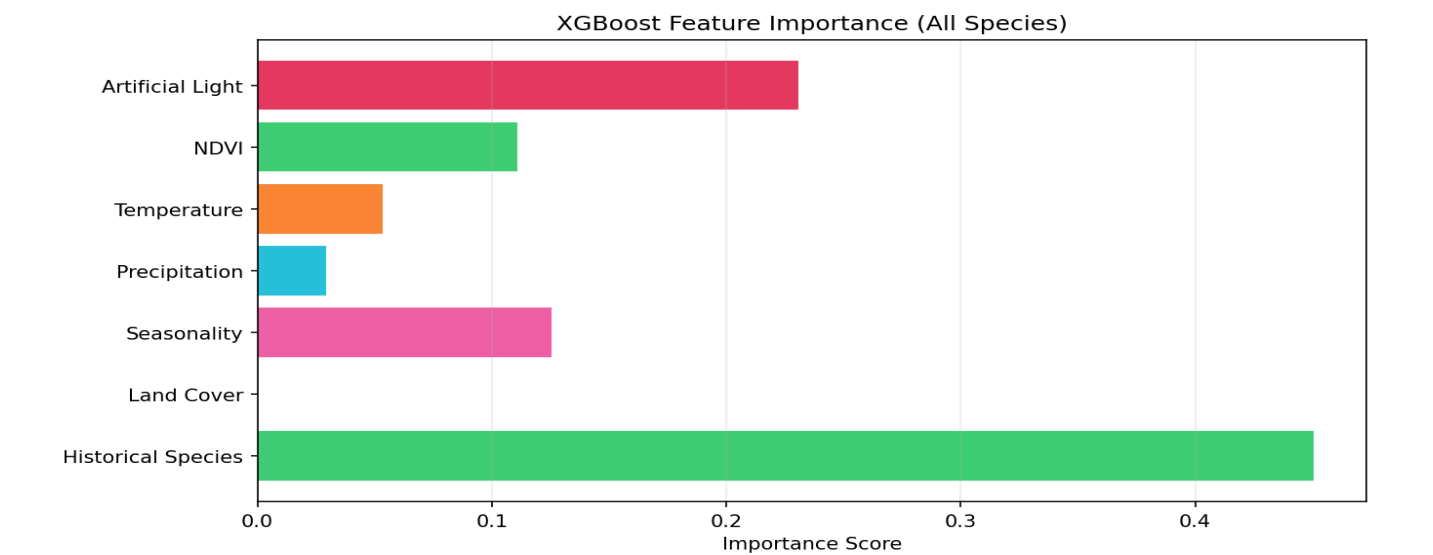
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Model Diagnostics View

KPI Cards

Metric	Value
RMSE	6.3924
MAE	4.0568
R2	0.7445

Chart: XGBoost Feature Importance (All Species)

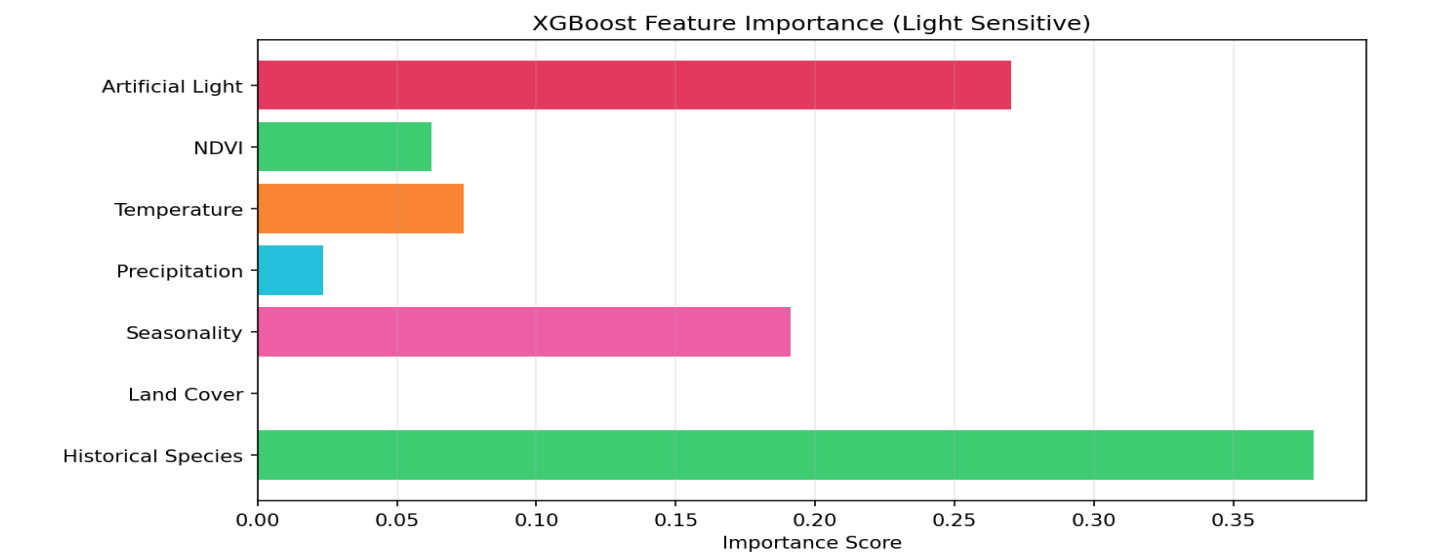


Numerical Values: XGBoost Feature Importance (All Species)

Feature	Importance
Artificial Light	0.230796
NDVI	0.110724
Temperature	0.053409
Precipitation	0.029266
Seasonality	0.125400
Land Cover	0.000000
Historical Species	0.450406

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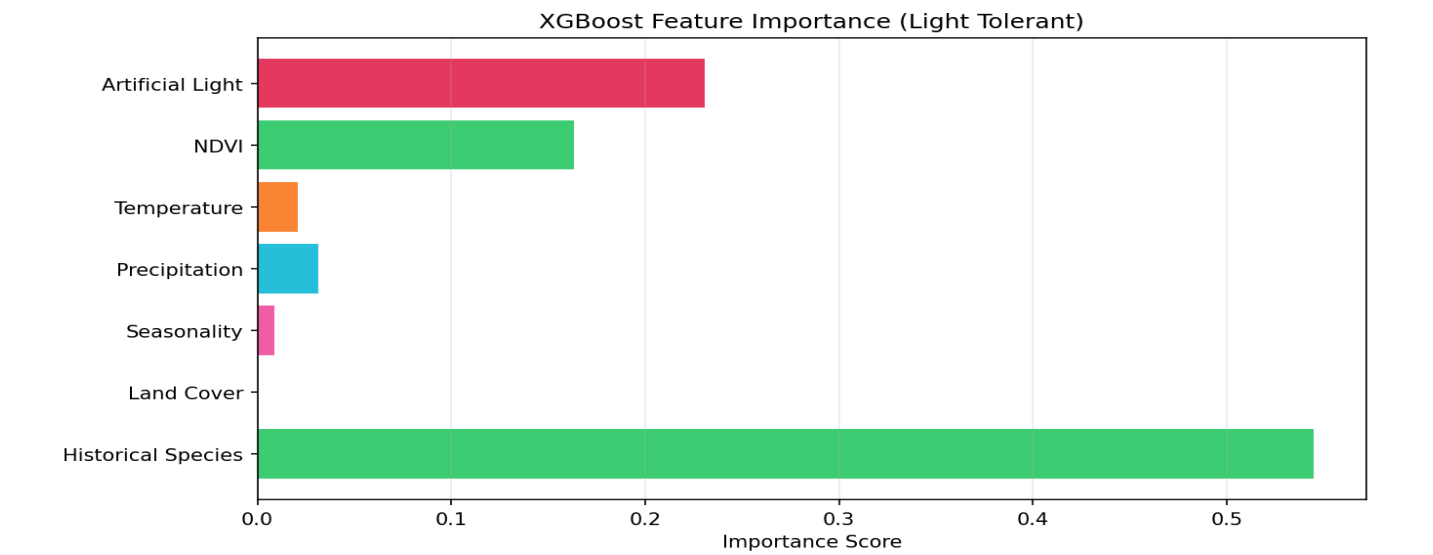
Chart: XGBoost Feature Importance (Light Sensitive)



Numerical Values: XGBoost Feature Importance (Light Sensitive)

Feature	Importance
Artificial Light	0.270363
NDVI	0.062340
Temperature	0.073999
Precipitation	0.023678
Seasonality	0.191019
Land Cover	0.000000
Historical Species	0.378602

Chart: XGBoost Feature Importance (Light Tolerant)



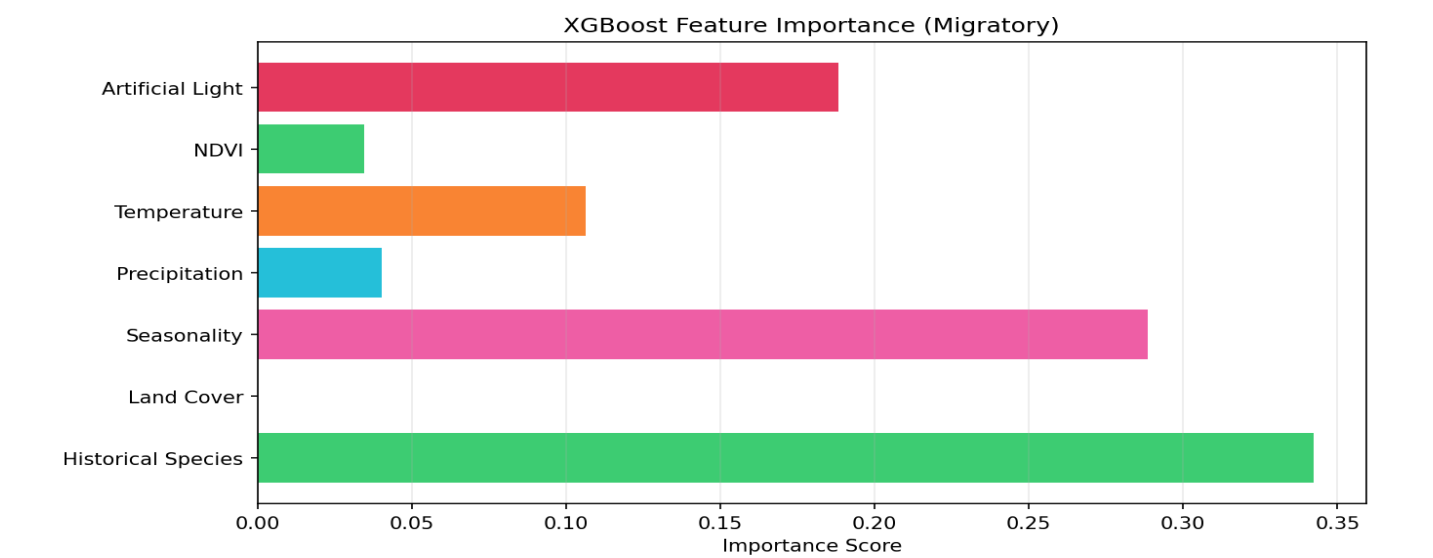
Numerical Values: XGBoost Feature Importance (Light Tolerant)

Feature	Importance
Artificial Light	0.230950
NDVI	0.163174
Temperature	0.020697

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Precipitation	0.031523
Seasonality	0.008989
Land Cover	0.000000
Historical Species	0.544666

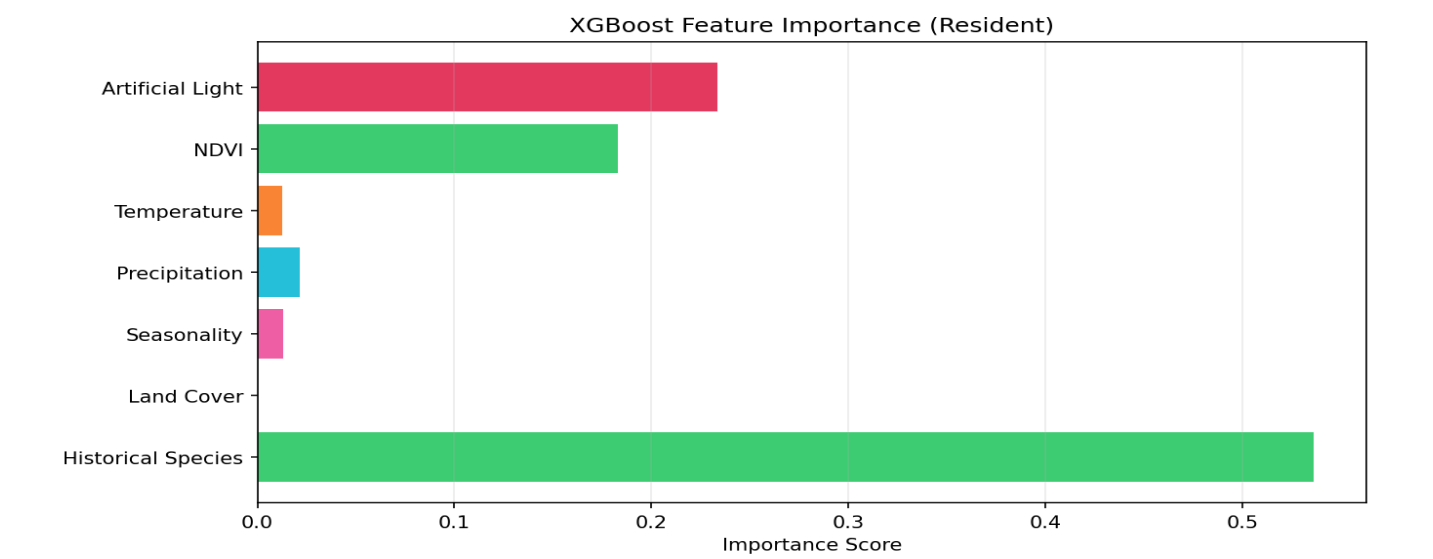
Chart: XGBoost Feature Importance (Migratory)



Numerical Values: XGBoost Feature Importance (Migratory)

Feature	Importance
Artificial Light	0.188284
NDVI	0.034494
Temperature	0.106226
Precipitation	0.040161
Seasonality	0.288632
Land Cover	0.000000
Historical Species	0.342202

Chart: XGBoost Feature Importance (Resident)

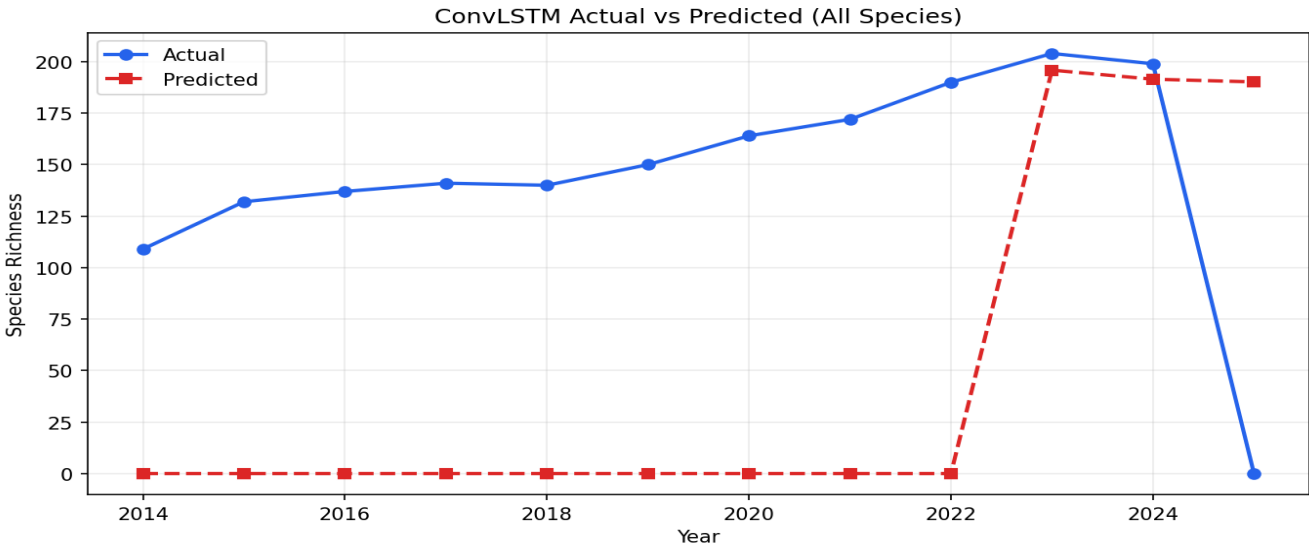


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Numerical Values: XGBoost Feature Importance (Resident)

Feature	Importance
Artificial Light	0.233587
NDVI	0.182887
Temperature	0.012712
Precipitation	0.021700
Seasonality	0.012960
Land Cover	0.000000
Historical Species	0.536153

Chart: ConvLSTM Actual vs Predicted (All Species)

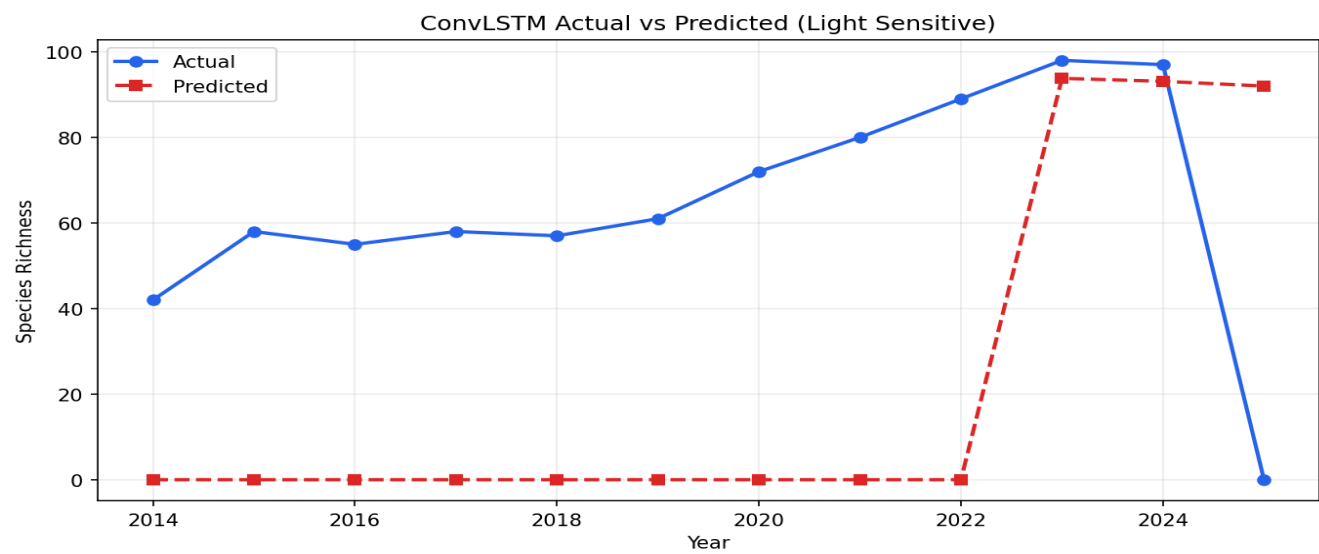


Numerical Values: ConvLSTM Actual vs Predicted (All Species)

Year	Actual	Predicted
2014	109.0000	0.0000
2015	132.0000	0.0000
2016	137.0000	0.0000
2017	141.0000	0.0000
2018	140.0000	0.0000
2019	150.0000	0.0000
2020	164.0000	0.0000
2021	172.0000	0.0000
2022	190.0000	0.0000
2023	204.0000	195.9000
2024	199.0000	191.5000
2025	0.0000	190.2000

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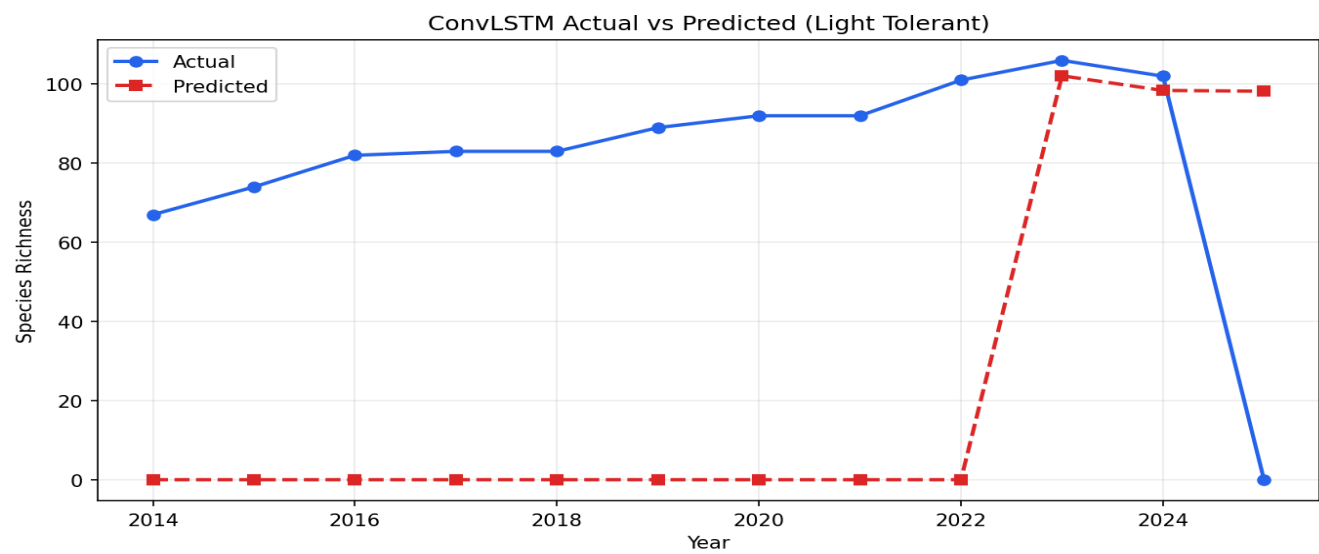
Chart: ConvLSTM Actual vs Predicted (Light Sensitive)



Numerical Values: ConvLSTM Actual vs Predicted (Light Sensitive)

Year		Actual	Predicted
	2014	42.0000	0.0000
	2015	58.0000	0.0000
	2016	55.0000	0.0000
	2017	58.0000	0.0000
	2018	57.0000	0.0000
	2019	61.0000	0.0000
	2020	72.0000	0.0000
	2021	80.0000	0.0000
	2022	89.0000	0.0000
	2023	98.0000	93.8000
	2024	97.0000	93.1000
	2025	0.0000	92.0000

Chart: ConvLSTM Actual vs Predicted (Light Tolerant)

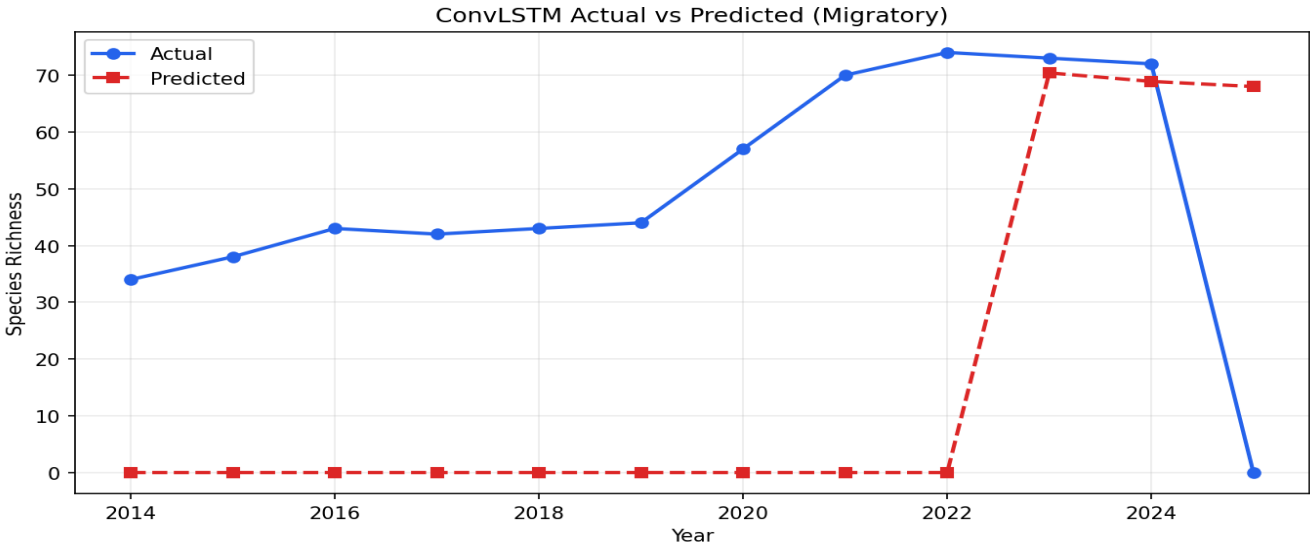


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Numerical Values: ConvLSTM Actual vs Predicted (Light Tolerant)

Year	Actual	Predicted
2014	67.0000	0.0000
2015	74.0000	0.0000
2016	82.0000	0.0000
2017	83.0000	0.0000
2018	83.0000	0.0000
2019	89.0000	0.0000
2020	92.0000	0.0000
2021	92.0000	0.0000
2022	101.0000	0.0000
2023	106.0000	102.1000
2024	102.0000	98.4000
2025	0.0000	98.2000

Chart: ConvLSTM Actual vs Predicted (Migratory)

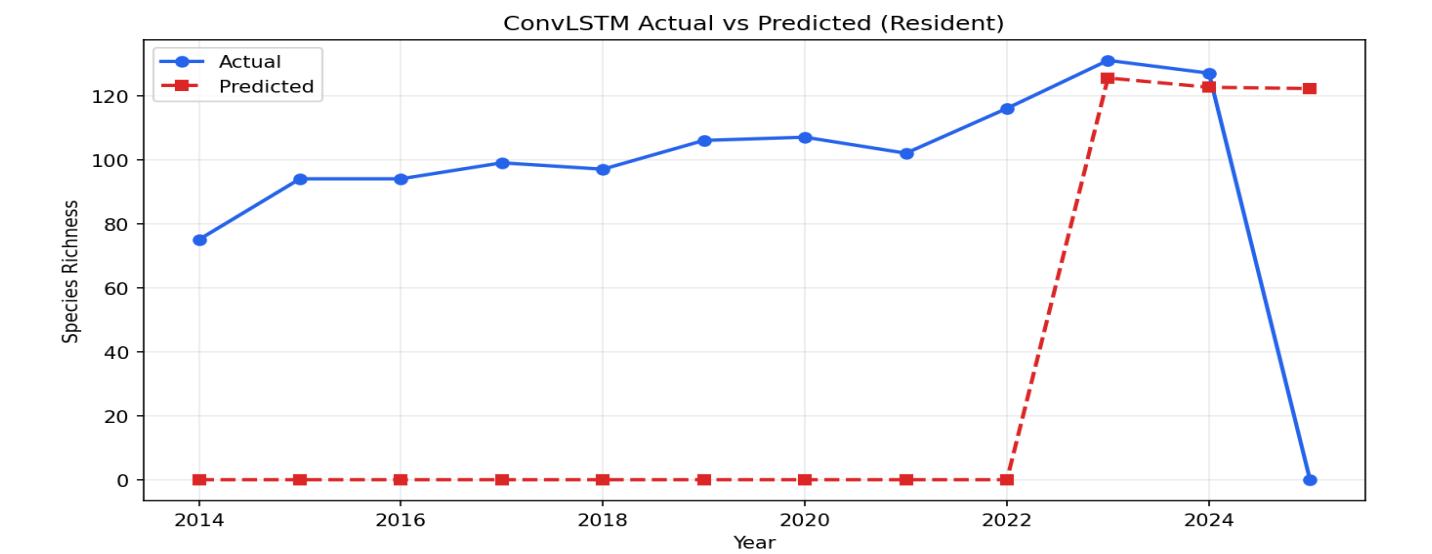


Numerical Values: ConvLSTM Actual vs Predicted (Migratory)

Year	Actual	Predicted
2014	34.0000	0.0000
2015	38.0000	0.0000
2016	43.0000	0.0000
2017	42.0000	0.0000
2018	43.0000	0.0000
2019	44.0000	0.0000
2020	57.0000	0.0000
2021	70.0000	0.0000
2022	74.0000	0.0000
2023	73.0000	70.4000
2024	72.0000	68.9000
2025	0.0000	68.0000

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Chart: ConvLSTM Actual vs Predicted (Resident)



Numerical Values: ConvLSTM Actual vs Predicted (Resident)

Year	Actual	Predicted
2014	75.0000	0.0000
2015	94.0000	0.0000
2016	94.0000	0.0000
2017	99.0000	0.0000
2018	97.0000	0.0000
2019	106.0000	0.0000
2020	107.0000	0.0000
2021	102.0000	0.0000
2022	116.0000	0.0000
2023	131.0000	125.5000
2024	127.0000	122.6000
2025	0.0000	122.2000

Systems Recommendations: KBA/PA Audit (Core)

Site	Type	Species	Sens.Cnt	Sens.%	Light	Eff.	Status
Manila Bay Beach Resort	PA	12	1	8.3	63.4	22.5	Critical
Luneta National Park	PA	29	6	20.7	78.9	39.3	Critical
Ninoy Aquino Parks and Wildlife Center	PA	41	10	24.4	48.2	48.3	At Risk
Manila Bay	KBA	92	35	38.0	11.6	53.4	At Risk
Las Piñas-Parañaque Wetland Park	PA	64	21	32.8	17.5	62.5	Moderate

Systems Recommendations: KBA/PA Audit (Environment)

Site	Mean NDVI	Max LST	Precip	Cells	Snapshot
Manila Bay Beach Resort	0.1952	36.69	209.70	45	2025-12
Luneta National Park	0.2073	37.42	27.01	8	2025-12
Ninoy Aquino Parks and Wildlife Center	0.3363	37.42	39.98	9	2025-12
Manila Bay	0.2370	37.42	4151.56	874	2025-12
Las Piñas-Parañaque Wetland Park	0.2970	36.23	198.71	30	2025-12